

Oliver Chang

oliver@oychang.com • +1-727-771-3641 • <https://oychang.com>

PROFESSIONAL EXPERIENCE

Modern Treasury, San Francisco, California

- Senior Software Engineer Mar 2024 – Mar 2025
 - Optimized GraphQL account search API using Postgres B-Tree GIN indexes and public product changes for a 100x decrease in latency for large customer edge cases
 - Updated edge routing service to use SQLite and S3 polling for dynamic routing changes, enabling non-technical users to onboard customers and enable multi-cell load balancing
 - Integrated merge queue on GitHub and Buildkite to avoid semantic merge conflicts with migrations and API contracts in a high velocity monolithic Ruby on Rails application

Sendbird, Inc., San Mateo, California

- Senior Backend Software Engineer Jun 2018 – Sep 2023
 - Led migration of core monolithic Python 2 chat backend to Python 3 with zero downtime and no customer side effects
 - Redesigned data retention architecture to use a backpressure-based event pool using multiple query strategies which lead to a 10% reduction in DB costs
 - Collaborated with Reddit to design and implement generalizable moderation features based on rules such as sliding window rate limits, a custom regular expression engine, and anti-flood detectors
 - Guided summer interns to complete projects for production features such as full-text search and moderation rule engine

Rice University, Computer Science Department

- Graduate PhD Researcher Aug 2016 – Mar 2018
 - Research focus on data-driven deep learning techniques for program synthesis
 - Taught labs as teaching assistant for graduate web development course (covering Git, HTML, CSS, Node, React)

Baker Institute, McNair Center for Entrepreneurship & Innovation

- Graduate Student Assistant May 2017 – Jan 2018
 - Created a parallel Extract, Transform, Load (ETL) system in Java to parse XML data from the US Patent and Trademark Office XML source files to create corpus for innovation research paper
 - Implemented a geoprocessing approach to measuring innovation via startup agglomeration for a published whitepaper

REConsole, Inc., Fort Lauderdale, Florida

- Software Engineering Intern Jan 2016 – May 2016
 - Automated a day's worth of manual engineering work by automating the data extract, transform, load (ETL) pipeline into a reproducible process built with Kubernetes, Apache Spark, Elasticsearch, and PostGIS
 - Worked with Lead UX Designer to systematically reduce geographical parsing errors or to find alternative methods of representing unreliable real estate transaction data

Florida International University, Electrical & Computer Engineering Department

- NSF REU Student, Mobile, Pervasive, and Autonomous Technologies Lab Oct 2015 – May 2016
 - Synthesized WiFi metadata, GPS location data, and qualitative user input in an Android app to perform indoor Simultaneous Localization and Mapping (SLAM) data collection
 - Won 2nd Place in 2015 University of Miami GIS Day with a use case, "GPS & WiFi Choke Point Analysis"

University of Illinois, Urbana-Champaign, Department of Computer Science

- NSF REU Student, Scientific Computing Group, Parallel Computing Institute Jun 2015 – Aug 2015
 - Optimized legacy Fortran77 MPI code using hardware-software features to attain an up to 20% speedup
 - Co-presented poster "Analyzing the Scalability of Nek5000" to the University of Illinois parallel community

National Aeronautics and Space Administration, Jet Propulsion Laboratory

- Summer Intern, Physical Oceanography Distributed Active Archive Center Jun 2014 – Aug 2014
 - Created a Python webservice for validating satellite data from third-party vendors using industry-standard specifications that were previously too hard to comply with or to understand
 - Poster featuring the design and merit of the work, "Improving Compliance for Earth Science Data Records," presented at the 2014 Meeting of the American Geophysical Union

EDUCATION

University of Miami, Coral Gables, Florida, USA

- B.S. in Computer Science & Geography Aug 2012 – May 2016
 - Mathematics Minor; Geospatial Technology Certificate
 - Magna Cum Laude with General and Departmental Honors

AWARDS & SCHOLARSHIPS

- Computer Aided Verification (CAV) Student Fellowship Jul 2017
- Outstanding Computer Science Undergraduate, University of Miami May 2016
- National Merit Scholarship Aug 2012 – May 2016

SKILLS

Python, Java, C, Unix, TensorFlow, Apache Spark, Amazon Web Services, Kubernetes, Docker, Elasticsearch, PostgreSQL, PostGIS, ESRI ArcGIS, MySQL, Redis, RabbitMQ, Django, Party Planning